

MAT 5283 001

Linear Algebra

Course Syllabus - Fall 2026



Course Information

Course Description: Prerequisite: MAT 2233 or an equivalent. A study of linear algebraic structures that may include linear transformations, inner product spaces, eigenvalues, Cayley-Hamilton theorem, similarity, the Jordan canonical form, spectral theorem for normal transformation and applications. This course has Differential Tuition. Course Fee: GS01 \$90.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 08/19/2026-12/11/2026

Campus: Main Campus

Location: [MH 3.03.10](#)

Time(s): MW 6:00 PM - 7:15 PM

Learning Goals

At the end of the course, you will be able to:

- Identify and work with basic algebraic structures
Outcome
Fundamental concepts concerning rings, domains, fields (including finite fields) will be mastered by students. Also, the Euclidean and Extended Euclidean Algorithm will be learned and applied to students in solving equations in more complex algebraic structures other than the real numbers.
- Identify and Apply Key results in Linear Algebra
Outcome
Students will learn and understand how to apply main results from Linear Algebra including: the Rank-Nullity Theorem for Matrices and Linear Transformations, Inner Product Spaces and the Gram-Schmidt Orthogonalization Process, Diagonalization and Similarity results concerning eigenvalues theorems and spectral theory, and lastly, elementary results on Multilinear Algebra.
- Apply results from Linear Algebra to other fields
Outcome
Linear algebra plays an important role in other fields like Biology, Chemistry, Physics, etc. The course will prepare students to apply and use linear algebra techniques in these fields. In

particular, examples from the areas of Error correction, Cryptography and Quantum Information will be included in the course.

Communicate with Me



Instructor Name: Henry Chimal-Dzul

Department
Mathematics

Office Location
FLN 4.02.46

Student Hours
Monday and Wednesday, 4:30PM to 6:00PM

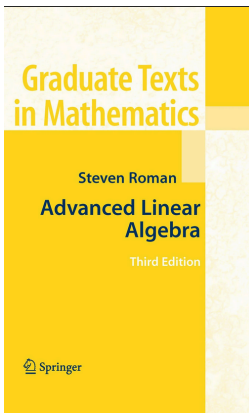
Email Address: henry.chimal-dzul@utsa.edu

Preferred Method of Communication
Email

About Me & My Teaching Philosophy

*I am excited to be part of your academic journey this semester. I have been working in the Academia for **7** years and have been at UTSA since **Fall 2024**.*

Course Materials



Advanced Linear Algebra

ISBN: 978-0-387-72828-5

Authors: Steven Roman

Publisher: Springer Verlag

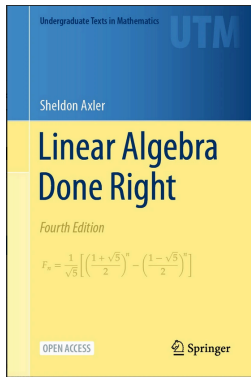
Publication Date: 2008

Edition: 3rd Edition

Required/Recommended/Optional: Required

Additional Information

<https://link.springer.com/book/10.1007/978-0-387-72831-5>

**Linear Algebra Done Right****ISBN:** 978-3-031-41025-3**Authors:** Sheldon Axler**Publisher:** Springer-Verlag**Publication Date:** 2024**Edition:** 4**Required/Recommended/Optional:** Recommended

Additional Information

<https://link.springer.com/book/10.1007/978-3-031-41026-0>

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Additional Course Information

Linear Algebra is a fundamental branch of mathematics and is regarded as the most powerful mathematical tool which has acquired enormous applications in diverse fields of studies viz. Basic Sciences, Engineering, Computer Science, etc. This course has been designed for advanced undergraduate students and first year graduate students. The subjects have been developed to cover topics ranging from vector spaces (of arbitrary dimension over finite and infinite fields) to an introduction to multilinear algebra. This has been done to include material that is fundamental for contemporary applications of linear algebra like Error-Correcting Codes, Cryptography, Quantum Computing and Quantum Algorithms. When possible, examples from these emerging fields are included in the development of the subjects.

Assessments and Assignments

Homework/Assignments

There are no formal, graded homework assignments for this course. Instead, **weekly practice problem sets** will be provided to help you master the material. While these are not turned in for a grade, working through them consistently is highly recommended to prepare for quizzes and exams.

Quizzes

Quizzes will be administered during the first **10 minutes** of class according to the schedule below.

- **Content:** Questions are based entirely on the lectures from the previous week.
- **Feedback:** Full solutions will be provided immediately after each quiz so you can track your understanding.
- **Grading Policy:** There are **12 total quizzes** throughout the semester. Your **two lowest scores will be automatically dropped**.

Quiz Schedule

Quiz Number	Date
Quiz 0 (Practice quiz, no graded is assigned)	Wednesday September 2, 2026.
Quiz 1	Wednesday September 9, 2026.
Quiz 2	Wednesday September 16, 2026.
Quiz 3	Wednesday September 23, 2026.
Quiz 4	Monday September 28, 2026.
Quiz 5	Wednesday October 7, 2026.
Quiz 6	Wednesday October 14, 2026.
Quiz 7	Wednesday October 21, 2026
Quiz 8	Wednesday October 28, 2026.
Quiz 9	Wednesday November 4, 2026.
Quiz 10	Wednesday November 11, 2026.
Quiz 11	Wednesday November 18, 2026.
Quiz 12	Wednesday December 2, 2026.

Midterm

The course will have a midterm administrated during lecture time on **Wednesday, September 30, 2026.**

- **Content:** All material covered from week 1 through week 5 (Monday August 24, 2026 to Wednesday September 23, 2026).
- **Structure:** The midterm is organized into three parts: True/False, Exercises concerning computations, concepts and Theorems, and proof-based exercises.
- **Feedback:** It will be provided the week after the midterm.

Final Exam

The final exam is **comprehensive**, meaning it will cover all material learned throughout the entire semester. It is scheduled in accordance with the official university calendar and will take place on **Monday, December 7, 2026, from 5:00 PM to 6:50 PM.** Please mark this date on your calendar now, as attendance is mandatory.

Activities and Grading

Activity	Quantity and Number of Points	Percentage
Quizzes	10 (20 points each; 200 points total)	20%
Midterm	1 (400 points)	40%
Final	1 (400 points)	40%
Total	1000	100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	
A	
A-	
B+	
B	
B-	
C+	
C	
C-	
D+	
D	
D-	
F	

Grade Distribution and Letter Grade

Course Guidelines for the Use of Generative Artificial Intelligence

Generative AI tool are encouraged to use during the course by students. It is expected from students to solve the sets of problems and ask for feedback to AI tools. That feedback should be used to improve students solutions and to improve their mathematical writing skills. **We always encourage students to attend office hours to discuss their solutions.**

Course Expectations & Policies

The lecture room is a place where you should feel free to engage in lively discussion about the course topic; don't be shy! But no course related interruptions should be kept to a minimum. In particular, you should turn off or switch to silent all phones, etc., before the start of class. If for some good reason you need to have your phone on during class, please mention it to me in advance.

All students are bound by the Honor Code pledge, which prohibits participation in or tolerance of academic dishonesty. In this course, students are encouraged to engage in collaborative discussion of assignments; however, the final written submission must represent each student's individual work. Any external sources of assistance (e.g., textbooks, software, or other materials) must be properly cited. Work submitted for the midterm and final examinations must be completed independently, in full compliance with the Honor Code and all stated examination policies.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Department Policies & Resources

Student Advocate

The student advocate is a person who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor; it is highly encouraged that you first reach out to the instructor with any issues to see if they can be resolved.

To reach the department's student advocate, please email mathematics@utsa.edu and include "Student Advocate" in the subject line of your email.

Tutoring Services

UT San Antonio provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213, MAT 1223, MAT 2213, MAT 2253, MAT 1313

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule for drop-in hours and course-specific targeted sessions will be provided by instructors through Canvas.
- **Location:** [Math Gym, Main Building 0.414](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, accommodations for pregnant students, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.
- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.

Ombudsperson

Ilse Rosales ilse.rosalesmartinez@utsa.edu

At UTSA, an ombudsperson is an informal, neutral, and confidential resource who helps faculty, staff, and students resolve disputes and concerns by providing listening, policy information, strategy development, and mediation services.